Illustration of the AIRS_spectralOLR diagnostics

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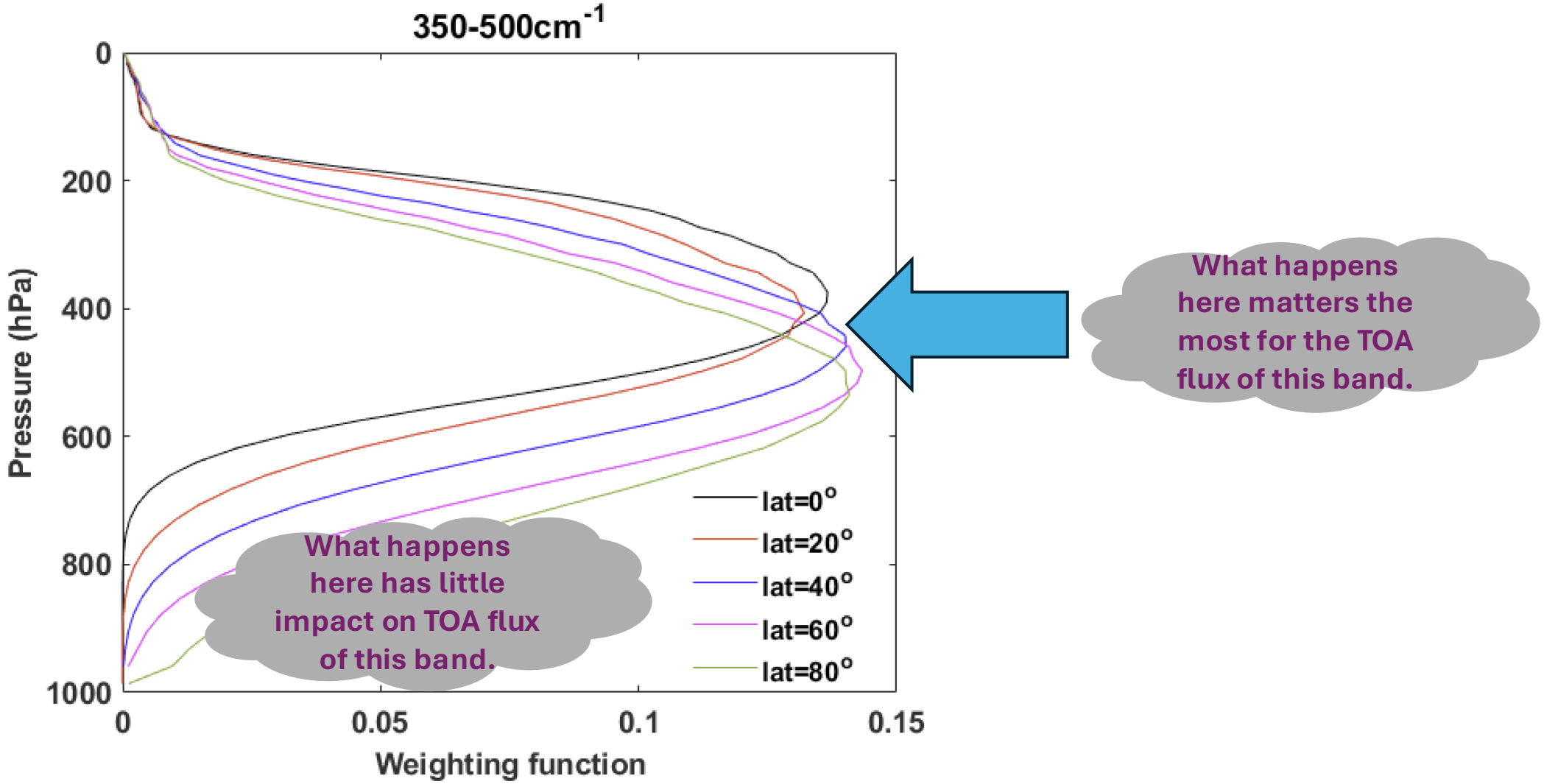
Physics background: why look these two bands in addition to OLR?

- They help you understand the ”origins” of biases in the OLR
  - Clear-sky 350-500 cm^{-1} vs. 820-980 cm^{-1} : upper troposphere vs. PBL
  - Cloud radiative forcing 350-500 cm^{-1} vs. 820-980 cm^{-1}: high clouds vs. all clouds
- It could reveal the compensating biases in the traditional OLR diagnostics
- Rule of thumbs
  - Any differences **350-500 cm^{-1}** are due to clouds/T/q in the upper and mid troposphere. **No low clouds or PBL T/q can affect it, regardless clear-sky or all-sky.**
  - Any differences in **820-980 cm^{-1} clear-sky flux** are primarily due **to T/q in PBL and surface temperature.**
  - Unlike the case of 350-500 cm^{-1}, **clouds at all levels** can affect **820-980 cm^{-1} all-sky flux**.

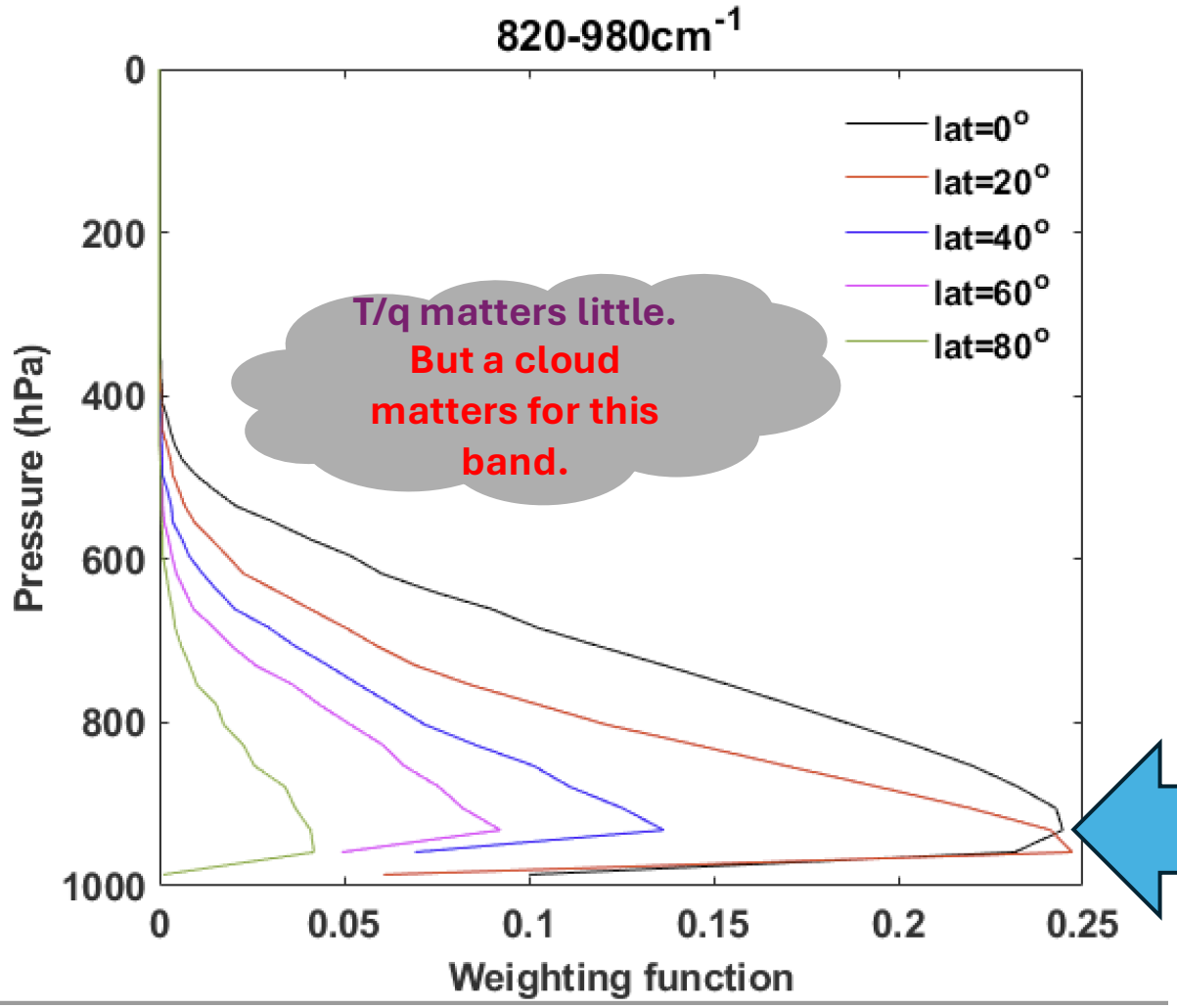
Physics background (II)

Clear-sky weighting function for band 350-500 cm-1 at different latitudes



Physics background (III)

Clear-sky weighting function for band 350-500 cm-1 at different latitudes



What happens here matters the most for clear-sky flux of this band

Sample explanations of the plot

- Based on plots from https://portal.nersc.gov/cfs/e3sm/chengzhu/tests/ex11-airs-spectral-olr/viewer/lat_lon/index.html

E3SM Diagnostics Package v3.1.1
Test: v3.LR.F20TR_spectral.pm-cpu
Reference: Observation and Reanalysis
Created: 2026-03-30 12:13:59

E3SM Energy Exascale Earth System Model

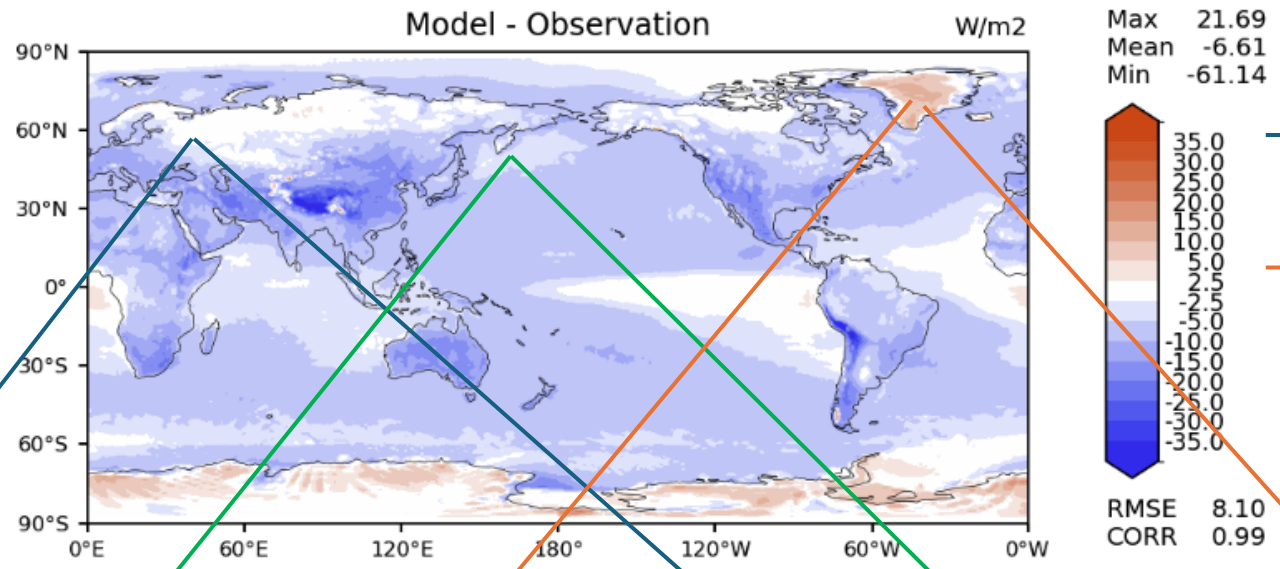
Latitude-Longitude contour maps

Jump To:

AIRS_specOLR

AIRS_specOLR	Description	ANN
FLUT global AIRS_specOLR	Upwelling longwave flux at top of model	ANN
FLUTC global AIRS_specOLR	Clearsky upwelling longwave flux at top of model	ANN
FLSU02 global AIRS_specOLR	all-sky LW spectral upward flux at TOA - band 02 350-500 cm-1	ANN
FLSU06 global AIRS_specOLR	all-sky LW spectral upward flux at TOA - band 06 820-980 cm-1	ANN
FLSUCLR02 global AIRS_specOLR	clr-sky LW spectral upward flux at TOA - band 02 350-500 cm-1	ANN
FLSUCLR06 global AIRS_specOLR	clr-sky LW spectral upward flux at TOA - band 06 820-980 cm-1	ANN
LWCF global AIRS_specOLR	TOA longwave cloud forcing	ANN
LWCF02 global AIRS_specOLR	TOA cloud forcing – band 02 350-500 cm-1	ANN
LWCF06 global AIRS_specOLR	TOA cloud forcing – band 06 820-980 cm-1	ANN
FLSU02_FRAC global AIRS_specOLR	Fractional contribution of band 02 to all-sky OLR	ANN
FLSUCLR02_FRAC global AIRS_specOLR	Fractional contribution of band 02 to clear-sky OLR	ANN
LWCF02_FRAC global AIRS_specOLR	Fractional contribution of band 02 to LW cloud forcing	ANN
FLSU06_FRAC global AIRS_specOLR	Fractional contribution of band 06 to all-sky OLR	ANN
FLSUCLR06_FRAC global AIRS_specOLR	Fractional contribution of band 06 to clear-sky OLR	ANN
LWCF06_FRAC global AIRS_specOLR	Fractional contribution of band 06 to LW cloud forcing	ANN

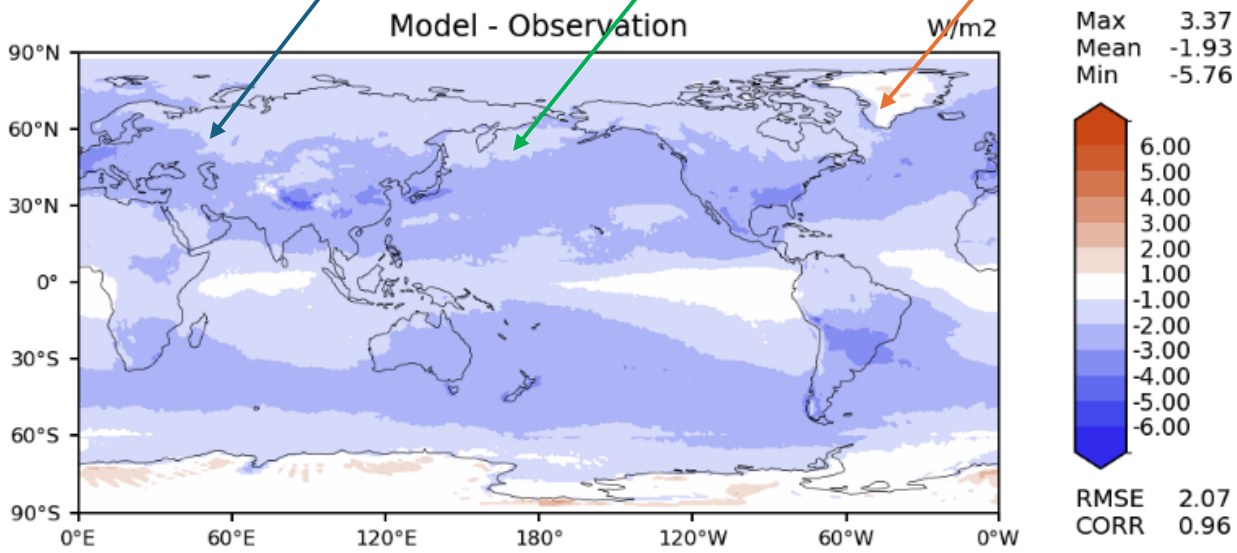
FLUTC (broadband)



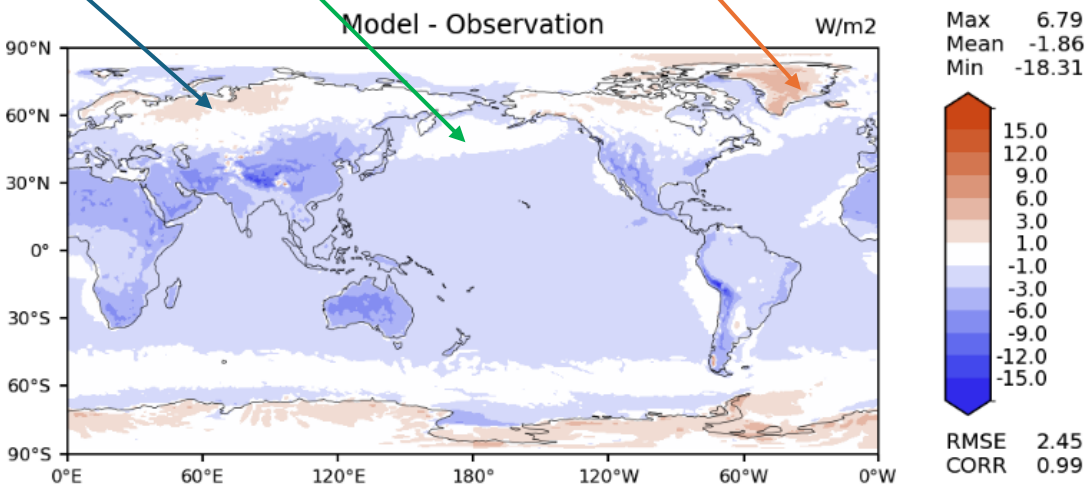
Compensating differences in northern europe

Greenland due to surface/PBL,not UT. So does Australia.

Band2: (upper troposphere)

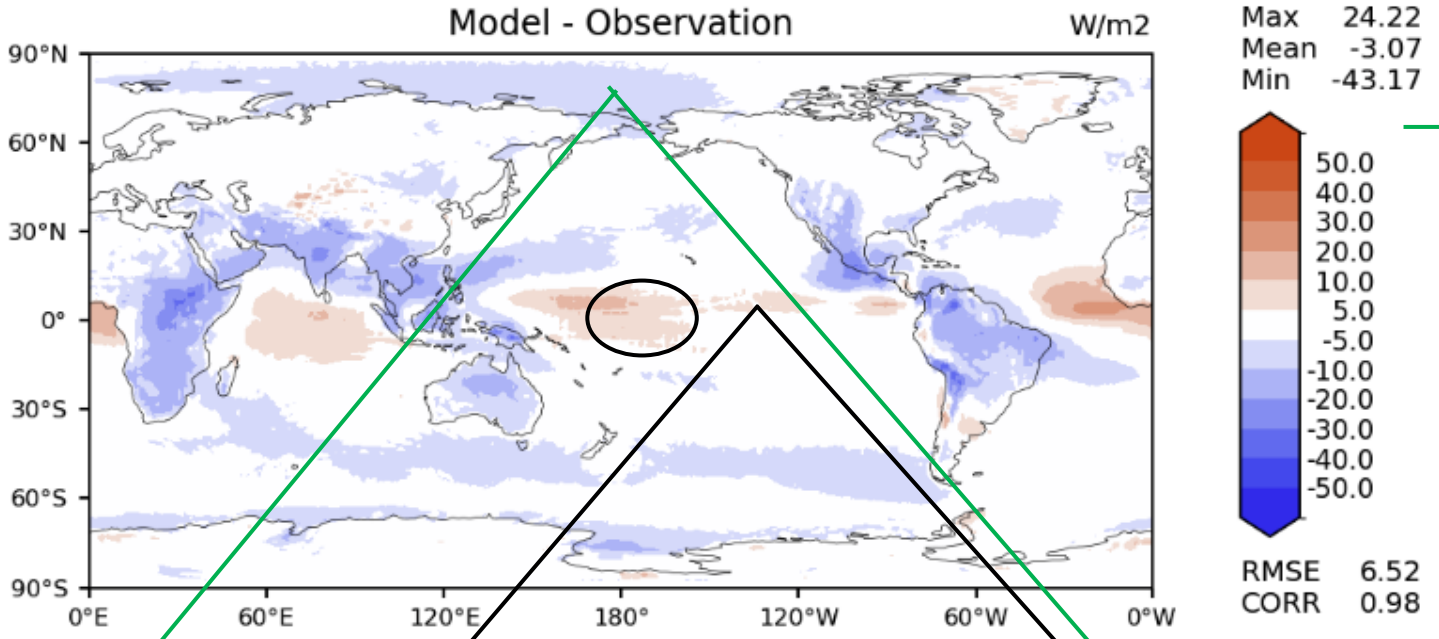


Band6: (PBL and surface)



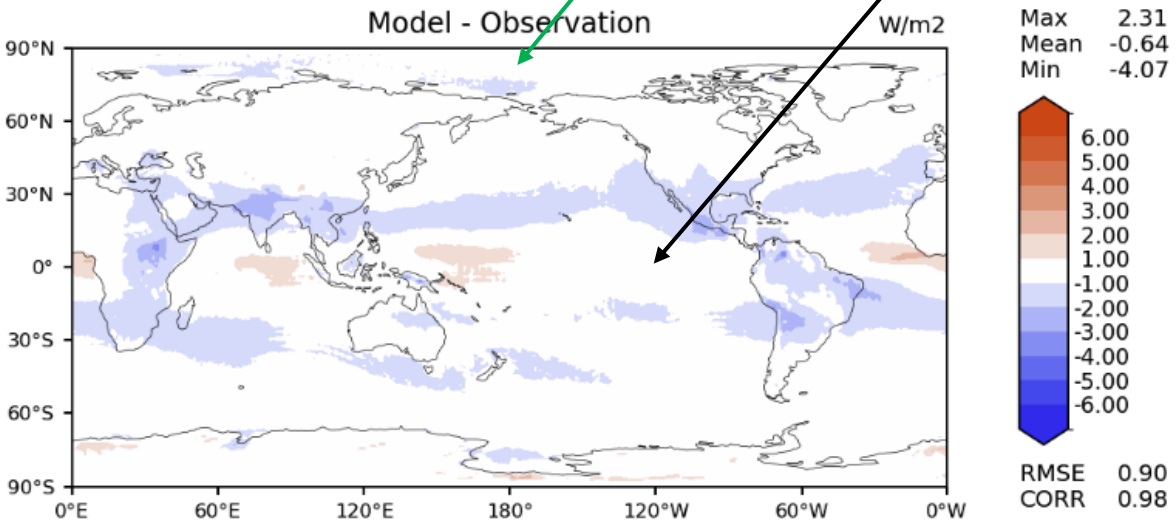
NW Pacific, largely due to UT

FLUT (broadband)

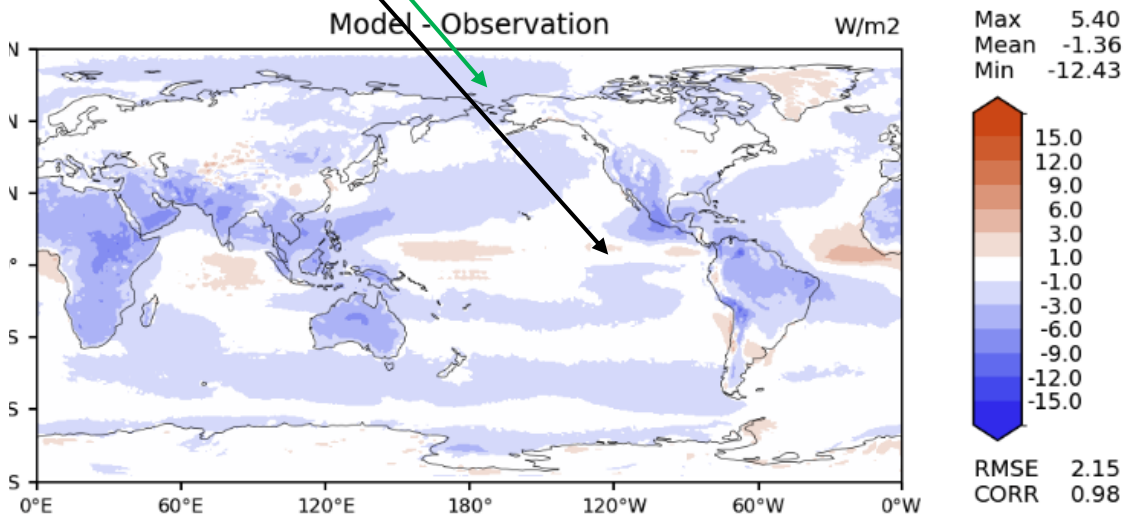


This must be low cloud, no showing in Band 2 and FLUTC

Band2: (upper troposphere T/q/clouds only)



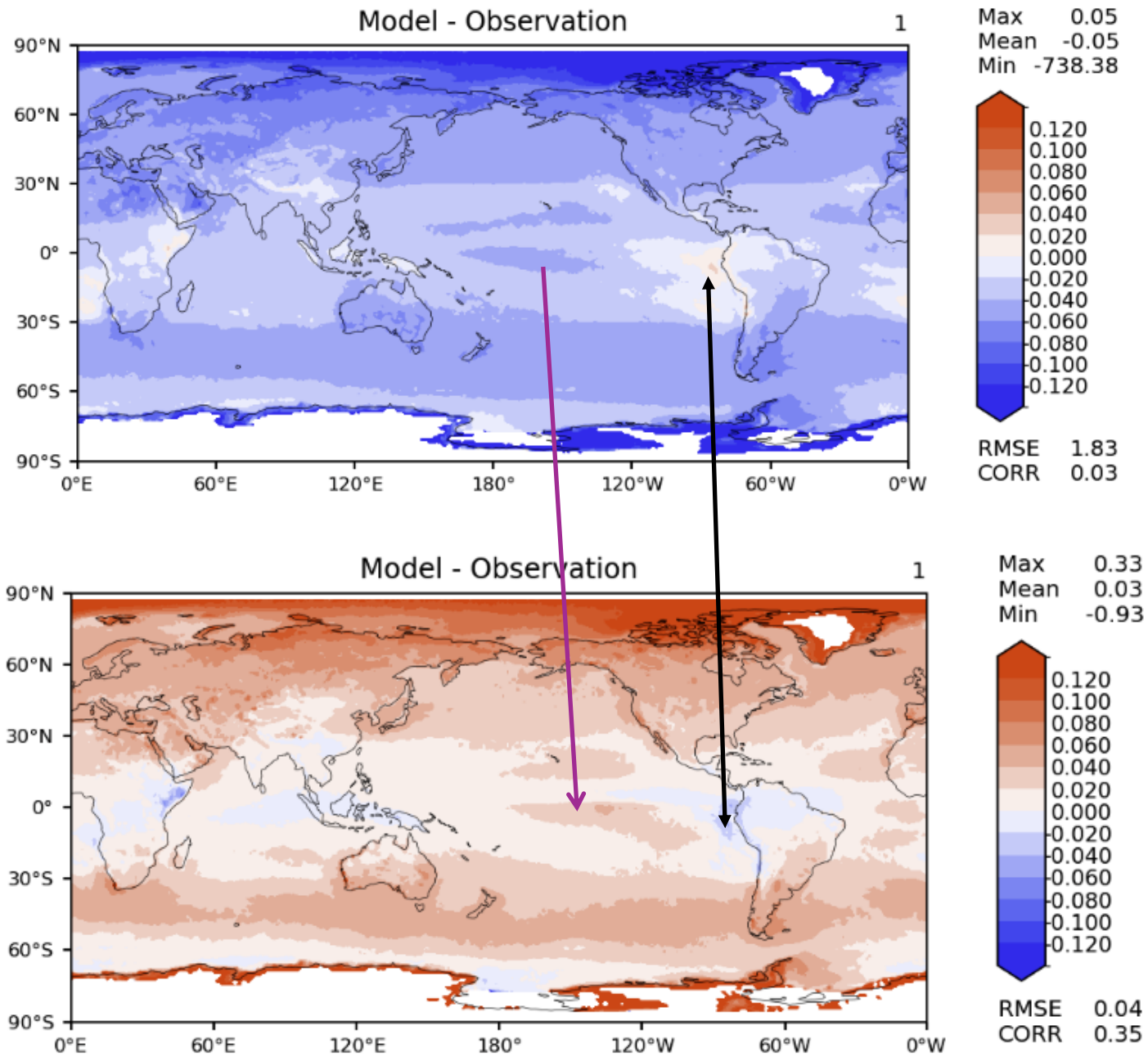
Band6: (PBL and surface, all clouds)



This bias is due to cloud, but the cloud is not as high as the cloud to its left (black circle)

Diagnosis using fractional contribution of LW Cloud forcing

- LW cloud forcing is largely decided by two factors: cloud fraction, and cloud top temperature
- Fractional contribution of a band to the broadband LW cloud forcing: More sensitive to cloud top temperature, less to cloud amount
- Rules of thumbs
  - If band2 fractional contribution is higher than observed, simulated cloud top is higher than observed. Correspondingly, band 6 fractional contribution is then lower than observed.
  - If band2 fractional contribution is lower than observed, simulated cloud top is lower than observed. Correspondingly, band 6 fractional contribution is then higher than observed.
  - **Band 2 vs. Band6 fractional contribution difference are opposite signs if it is due to cloud height bias**
- Note: (1) these rules are more useful in the extra-polar regions. Frequent inversion in the polar regions makes the rule invalid. (2) In reality, simulated clear-sky biases can affect the fractional contributions discussed here too.



Fractional contribution of band2 to LW Cloud forcing

Cloud top lower than obs

Cloud top higher than obs

Fractional contribution of band6 to LW Cloud forcing

Differences in such fractional contributions from two bands are usually opposite to each other